Introduction to Machine Learning

https://drive.google.com/file/d/1M84oM7MTgp6iCqG4MieO9nFL-dUwaW-s/view
[+ HTML notes]

📌 What is Machine Learning (ML)?

- "Field of study that gives **computers the ability to learn without being explicitly programmed**."

💡 Why Do We Need ML?

ML is useful when:

- ❌ **Human expertise does not exist**
  → Example: Mars rover navigation
- 🤯 **Humans cannot explain** their **expertise**
  → Example: Speech recognition
- 📊 **Data is too large**
  → Example: Data mining
- 🔮 Need to **predict new data**
  → Example: Stock prices
- 🔁 **Tasks improve with practice**
  → Example: Robot path planning

📌 Inductive Learning

Goal: Learn a function from examples.

- Given:
  - Training data:

$$\{(x_1,y_1),\ (x_2,y_2),\ \dots,\ (x_n,y_n)\}$$

- Learn:
  - Function **f: x → y**
- Find hypothesis:
  - **h ≈ f**

✔️ Key Terms

- **Hypothesis (h):** Learned model
- **Consistent: h(x) = f(x) for all training data**

📈 Learning Process

Training Data → Learn Model → Predict new x

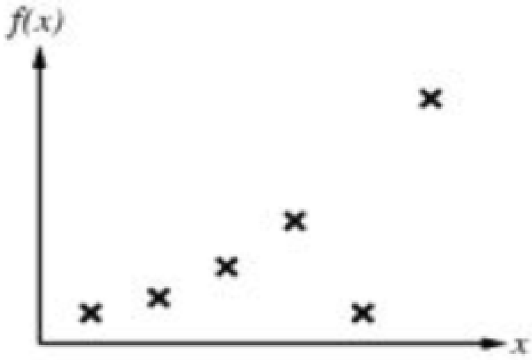
› example: [curve fitting]

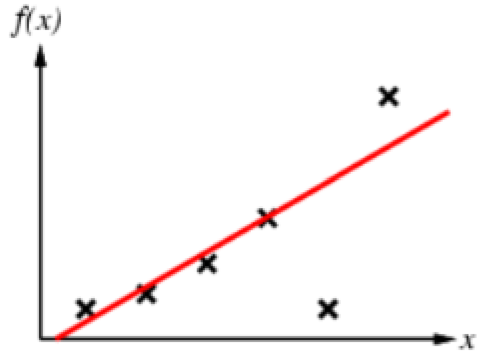
Curve fitting: Finding a mathematical function (**curve**) that **best matches** the given data points.

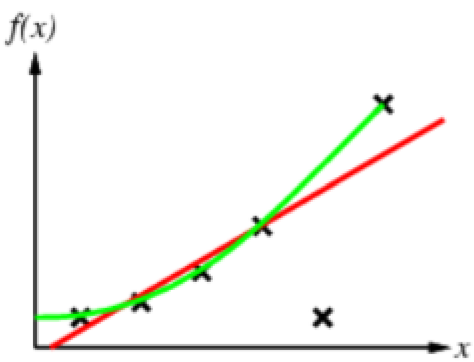
- You have data points: (x, y)
- You try to draw a curve/function that **captures the pattern**

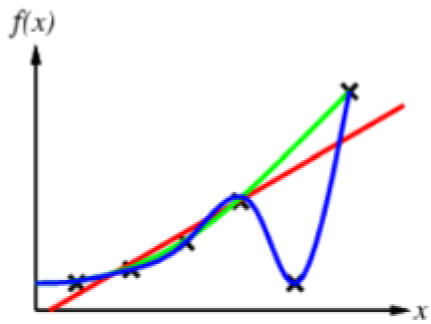
👉 In ML terms:

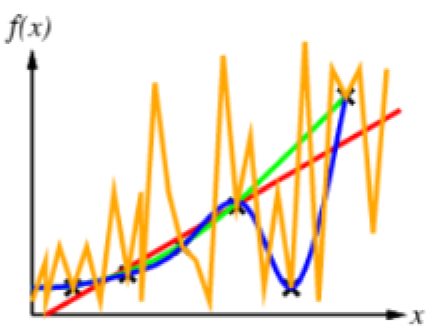
- **True function** = **f(x)** (unknown)
- Learned model = **h(x)**
- Goal: **h(x) ≈ f(x)**







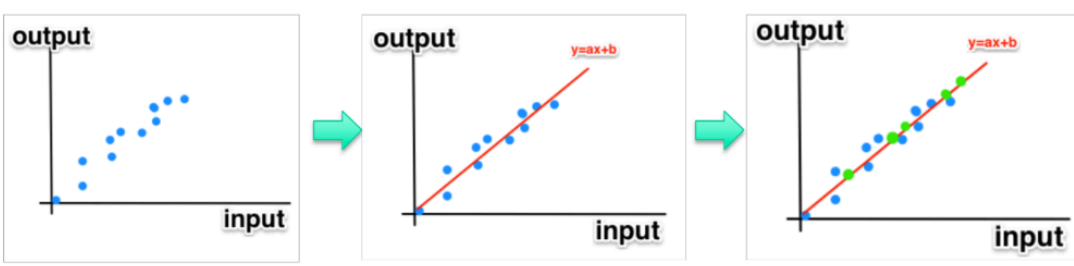




1. **Linear Hypothesis (Red Line)**: This is a simple linear model. It captures the general upward trend but does not pass through every point. It might be too simple, potentially leading to **underfitting**.
2. **Quadratic/Polynomial Hypothesis (Green Curve)**: This model is more flexible. It follows the data points more closely than the straight line, suggesting a more nuanced relationship between x and $f(x)$.
3. **Complex Polynomial (Blue Curve)**: This curve is **consistent** because it passes through every single data point. However, it starts to look "wiggly." This is often a sign of **overfitting**, where the model learns the noise in the data rather than the actual trend.
4. **Extreme Overfitting (Orange Jagged Line)**: This final stage shows a highly complex hypothesis. While it hits every point, it is clearly a poor representation of the likely real-world function. It has "memorized" the training set but would fail to predict new, unseen data points accurately.

› What is meant by learning?

- writing algorithms that **identify patterns within data** rather than following **strictly hard-coded instructions**
- objective of these algorithms is to create a **statistical model** that acts as a **mathematical approximation** of the provided data



Data from Past Experiences

Calculating a model

Estimating the output for new input values

📌 Challenges of Machine Learning

1. High Dimensionality

- **More features** → more **complexity**
- Needs:
  - **More memory**
  - **More computation**
- ⚠️ **Causes overfitting**

👉 Example: DNA data

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2. Choice of statistical Model

- Wrong model → poor results
- **Too simple** → **Underfitting**
- **Too complex** → **Overfitting**

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3. Noise & Errors

- **Gaussian Noise:** Normal random variation
- **Outlier:** Very different data point / distant from the rest of the data
- **Inlier:** local outlier
- Can be due to measurement errors [**human error**]

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4. Insufficient training Data

- **Not enough data** → **poor learning**

👉 Also important:

- **Feature Extraction**
  - Reduce data into useful representation

📌 Types of Machine Learning

TYPE	TRAINING DATA	GOAL	CATEGORIES
Supervised	$\{(x_1,y_1), \ldots, (x_n,y_n)\}$	Learn $f: x \to y$	Regression, Classification
Unsupervised	$\{x_1, \ldots, x_n\}$ only	Find structure in inputs	Clustering, Dimensionality Reduction
Reinforcement	Delayed reward signal	Learn a policy (sequence of outputs)	Policy-based, Value-based

Type	Data	Goal	Examples
Supervised	(x, y)	Learn mapping	Regression, Classification
Unsupervised	x only	Find structure	Clustering
Reinforcement	Reward signal	Learn actions	Robotics

📌 Supervised Learning

- Supervised learning involves teaching a computer using data that already contains the "answers."
- **Training Data**: The dataset contains **input vectors x** paired with their **corresponding target** or **output vectors y**
- **Goal**: To find a mathematical model that can accurately predict the output y for any new, unseen input x

1. Regression

used when the outcome you **want to predict** is a **specific number**.

- Output: **Continuous (numeric)**

👉 Example:

- **Input**: Area in m^2.
- **Output**: Price in thousands of dollars.

> Area = 80 m^2 → Price ≈ 155K

2. Classification

> used to identify **which group or category** a piece of **data belongs to**

- Output: **Discrete labels**

👉 Example:

Computer → "Good" or "Bad"

Feature	Regression	Classification
Output Goal	Estimate a numerical response	Identify group membership
Visual Goal	**A line passing through points**	**A boundary separating groups**

📌 Unsupervised Learning

Goal ⇒ Find hidden structure (no labels)

- the computer explores the data to find its own logic.
- **Training Data**: Contains only input vectors without any target labels

Clustering Types - Grouping by Similarity

Clustering aims to **group data points** that are **similar to each other**

- **Connectivity-based**
  → **Nearby** points **grouped**
- **Centroid-based (K-Means)**
  → grouped around a **central point** (**centroid**).
- **Distribution-based**
  → **Same probability distribution**
- **Density-based**
  → **Dense regions form clusters**

Dimensionality Reduction

- **Reduce number of features**

Uses

- **Visualization**
- **Storage reduction**
- **Feature extraction**
- Makes **complex data easier** to an**a**lyze.

📌 Reinforcement Learning (RL)

- Agent learns by **trial and error**

Process

1. **Take action**
2. **Get reward**/**penalty**
3. **Update strategy** (policy)

💡 Key Idea

- **Good paths** → **higher weights**
- **Bad paths** → **lower weights**
- Over time → optimal path

📌 Model Selection

- Choose **simplest model** that works

Overfitting

- **Too complex model**
- **Memorizes training data**

✔ **High training accuracy**
❌ **Low test accuracy** → fails on test data

Underfitting

- **Too simple model**

❌ **Poor training** accuracy
❌ **Poor test** accuracy

📌 Data Splitting

A data set is split into the following:

1. Training Set

- **Train model**

2. Validation Set

- **Tune parameters**

3. Test Set

- **Final evaluation (use only once)**

📌 K-Fold Cross Validation

Steps

1. **Split data into K parts**
2. **Train** on **K-1** folds
3. **Test** on **1 fold**
4. **Repeat K times**

Formula

$$CV = \frac{1}{K} \sum_{k=1}^{K} Score_k$$

📌 Confusion Matrix

	Predicted +	Predicted -
Actual +	TP	FN
Actual -	FP	TN

📊 Metrics

Accuracy

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

Precision

$$Precision = \frac{TP}{TP+FP}$$

Recall

$$Recall = \frac{TP}{TP+FN}$$

F1 Score

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision+Recall}$$